

Data Definition Language (DDL):

1. Create a table named `Employees` with columns for `EmployeeID`, `FirstName`, `LastName`, `Department`, and `Salary`.

```
CREATE TABLE Employees (  
    EmployeeID INT PRIMARY KEY,  
    FirstName VARCHAR(50),  
    LastName VARCHAR(50),  
    Department VARCHAR(50),  
    Salary DECIMAL(10,2)  
);
```

2. Add a column named `HireDate` to the `Employees` table.

```
ALTER TABLE Employees  
ADD HireDate DATE;
```

3. Rename the column `Department` in the `Employees` table to `Dept`.

```
ALTER TABLE Employees  
RENAME COLUMN Department TO Dept;
```

4. Remove the column `Salary` from the `Employees` table.

```
ALTER TABLE Employees  
DROP COLUMN Salary;
```

5. Create a new table named `Departments` with columns for `DeptID` and `DeptName`.

```
CREATE TABLE Departments (  
    DeptID INT,  
    DeptName VARCHAR(50)  
);
```

6. Add a primary key constraint to the `DeptID` column in the `Departments` table.

```
ALTER TABLE Departments  
ADD PRIMARY KEY (DeptID);
```

7. Create a foreign key constraint on the `Employees` table that references the `DeptID` column in the `Departments` table.

```
ALTER TABLE Employees  
ADD DeptID INT;  
  
ALTER TABLE Employees  
ADD CONSTRAINT fk_employee_department  
FOREIGN KEY (DeptID)  
REFERENCES Departments(DeptID);
```

Data Manipulation Language (DML):

1. Insert a new employee into the `Employees` table with the following details: EmployeeID = 101, FirstName = "John", LastName = "Doe", Dept = "IT", Salary = 60000, HireDate = '2024-05-14'.

```
INSERT INTO Employees  
(EmployeeID, FirstName, LastName, Dept, Salary, HireDate)  
VALUES  
(101, 'John', 'Doe', 'IT', 60000, '2024-05-14');
```

2. Update the salary of employee with `EmployeeID` 101 to 65000.

```
UPDATE Employees  
SET Salary = 65000  
WHERE EmployeeID = 101;
```

3. Delete the employee with `EmployeeID` 101 from the `Employees` table.

```
DELETE FROM Employees
```

```
WHERE EmployeeID = 101;
```

Constraints:

1. Create a table named `Students` with columns for `StudentID`, `FirstName`, `LastName`, and `DateOfBirth`.

```
CREATE TABLE Students (  
    StudentID INT,  
    FirstName VARCHAR(50),  
    LastName VARCHAR(50),  
    DateOfBirth DATE  
);
```

2. Add a NOT NULL constraint to the `FirstName` column in the `Students` table.

```
ALTER TABLE Students  
MODIFY FirstName VARCHAR(50) NOT NULL;
```

3. Add a UNIQUE constraint to the `StudentID` column in the `Students` table.

```
ALTER TABLE Students  
ADD CONSTRAINT uq_student_id UNIQUE (StudentID);
```

4. Create a CHECK constraint on the `DateOfBirth` column in the `Students` table to ensure that dates are not in the future.

```
ALTER TABLE Students  
ADD CONSTRAINT chk_date_of_birth  
CHECK (DateOfBirth <= CURDATE());  
  
ALTER TABLE Students  
ADD CONSTRAINT chk_date_of_birth  
CHECK (DateOfBirth <= '2026-08-17');
```

5. Add a DEFAULT constraint to the `LastName` column in the `Students` table, setting it to "Doe" by default.

```
ALTER TABLE Students
```

```
ALTER COLUMN LastName SET DEFAULT 'Doe';
```